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A Comparative Study of RAG Techniques for Generating Image Captions

Phase 1 Technical Report

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Abstract

Automatic image captioning is an important task in computer vision and natural language processing. However, vision-language models may omit objects, describe objects that are not present, or fail to accurately represent visual attributes and relationships. Retrieval-Augmented Generation (RAG) offers a potential solution by providing relevant visual and textual information to support caption generation.

This project aims to investigate how different RAG techniques influence the quality, relevance, and visual grounding of generated image captions using the MS COCO 2017 dataset. The main motivation is to determine whether incorporating retrieved evidence can improve caption accuracy, reduce hallucinations, and provide a better balance between generation quality and computational cost.

The first phase focuses on building a reproducible data preparation and retrieval framework. It includes a knowledge base of 10,000 images and 50,000 captions, alongside separate validation and test sets of 100 images each. The pipeline covers image and caption preprocessing, metadata extraction, embedding generation, FAISS indexing, exploratory data analysis, and anti-leakage validation. Five retrieval configurations based on CLIP, SigLIP, and DINOv2 were evaluated using CIDEr scores on the validation set. SigLIP image-to-text achieved the highest validation CIDEr score of 0.7223 and was selected as the best-performing configuration.

At this stage, the evaluation focuses on retrieval performance rather than generated captions.

Keywords: MS COCO, image captioning, retrieval-augmented generation, CLIP, DINOv2, SigLIP, Sentence-BERT, FAISS, data leakage.

1 Introduction and Problem Statement

The objective of this project is to study retrieval-based methods for generating image captions. Image caption generation normally requires an image encoder, a language-generation model, and an image-text interface; in a retrieval-augmented formulation, the system first retrieves relevant images or captions from an external knowledge base and then uses that evidence as conditioning context. Phase 1 is a data-engineering and benchmarking phase rather than an implementation of the final caption generator. It therefore prepares a common, verified corpus and the retrieval infrastructure that later groups can use consistently.

Phase 1 builds a clean, shared image-caption dataset, prevents data leakage, selects the best encoder using validation data, and establishes a retrieval baseline that finds similar images or captions without generating new captions.

The scientific deliverable is not a claim that Phase 1 itself generates captions. Instead, it supplies a retrieval baseline using nearest-neighbour candidates and compares candidate captions with reference captions. This distinction is reflected in the experimental vocabulary: an image-to-image configuration retrieves KB images and then uses an associated caption, while an image-to-text configuration retrieves a KB caption vector directly. Both can serve as precursor evidence for later retrieval-augmented generation.

2 Dataset Description

The project uses MS COCO 2017 [3]. COCO provides images with object-instance annotations and five or more human-authored captions associated with many images. This report is restricted to the implemented Phase 1 sizes documented in Table 1. The KB contains 10,000 images and 50,000 retained captions, whereas validation and test each contain 100 images and 500 raw reference-caption rows.

Table 1: Release composition.

Split	Source	Images	Captions/references
Knowledge base	COCO train2017	10,000	50,000 KB captions
Validation	COCO train2017	100	500 references
Test	COCO val2017	100	500 references
Total		**10,200**	**51,000**

Table 1 summarizes the dataset composition. The KB and validation sets come from separate selections of COCO train2017, while the test set comes from COCO val2017.

Duplicate removal is important because repeated images can cause data leakage, bias evaluation results, and reduce retrieval diversity. It also avoids unnecessary storage and embedding computation. This is particularly important for SigLIP and FAISS, where duplicate images could occupy multiple positions in the retrieval results.

3 Dataset Splitting and Stratification

3.1 Sampling order and eligibility

Sampling follows this order: test, validation, then KB. Test images come from COCO val2017, while validation and KB images come from train2017. Validation IDs are excluded from KB selection.

Images must meet the minimum object annotation count, contain at least one caption, and have a shorter side of at least 200 pixels. Sampling uses a fixed global seed of 42, with separate seeds for each split.

Images without object annotations or below the required size are excluded because they do not meet the pipeline requirements.

3.2 Stratification algorithm

Validation and test are stratified by scene complexity, image context, object interactions, and class coverage. Complexity is based on the number of non-crowd object annotations:

$$\text{simple} = 1\text{-}2, \quad \text{medium} = 3\text{-}5, \quad \text{complex} = 6\text{-}9, \quad \text{crowded} \geq 10.$$

First, a greedy selection step prioritizes images containing rare COCO categories and covering the largest number of remaining categories. Seeded tie-breaking ensures reproducible selection. The complexity targets are 25% simple, 30% medium, 30% complex, and 15% crowded. Rounding preserves the requested sample size. Overfilled groups are rebalanced, and remaining images are assigned to medium complexity.

An image is considered to contain an interaction if it has a person and another class, at least two people, or an animal together with another supercategory. At least 30% of each evaluation sample must contain interactions. Selection is performed within complexity groups, rotating between indoor, outdoor, and mixed contexts. Interaction images are prioritized until the 30% target is reached.

Table 2. Observed stratification of validation and test.

Split	Simple	Medium	Complex	Crowded	Indoor	Outdoor	Mixed	Interaction	Classes	Min./class
Validation	25	30	30	15	35	35	30	69 (69%)	80	1
Test	25	30	30	15	37	33	30	61 (61%)	80	1

Table 2 is referenced below in the context of representativeness. Both samples attain the exact complexity quotas. Their interaction fractions exceed the 30% minimum, and both cover all 80 COCO categories present in the implementation. Validation coverage is slightly more balanced between indoor and outdoor contexts, whereas test contains 37 indoor, 33 outdoor, and 30 mixed images. The context labels are saved sampling outputs, rather than claims of a separately enforced context-count design.

The KB remains a naturally distributed sample subject to its category floor. Its split summary contains at least 66 images for each class, which is above the configured floor of 60; this distinction is intentional, because the KB should reflect retrieval prevalence rather than the equally represented evaluation quotas.

4 Exploratory Data Analysis

4.1 Dataset-scale and annotation profile

The EDA summarizes the 10,200 images, 51,000 captions, and 80 object categories across the dataset. The KB contains 80,498 object instances. The average number of objects per image is 8.05 for KB, 7.74 for validation, and 6.77 for test. The median is 5 for KB and validation, and 4 for test. The maximum number of objects in a single image is 67, 65, and 62, respectively, as shown in Table 3.

Table 3. Object-content summary.

Split	Mean objects/image	Median	p90	Maximum	Mean categories/image
KB	8.05	5	19	67	3.25
Validation	7.74	5	18	65	3.41
Test	6.77	4	15	62	3.30

Table 3 explains why the validation and test captions can be expected to contain more compact information than individual estimates from object counts alone: the split means differ, yet medians indicate similarly constrained central scenes. Figure 1 presents the most frequent KB classes.

Person is the most frequent category in the KB, followed by classes such as car, chair, book, and bottle. These statistics help describe the dataset and identify common object categories.

The difference between KB and test scene complexity is statistically significant ($\chi^2 = 18.3304$, $p = 0.000376$). This difference is expected because the KB keeps its natural distribution, while the test set was deliberately balanced using the predefined stratification targets.

4.2 Spatial relations and layout

Figure 3 presents geometric relation frequencies. The repository derives relations by normalized bounding-box geometry, not semantic scene understanding. Left-of and right-of are complementary pair counts and occur

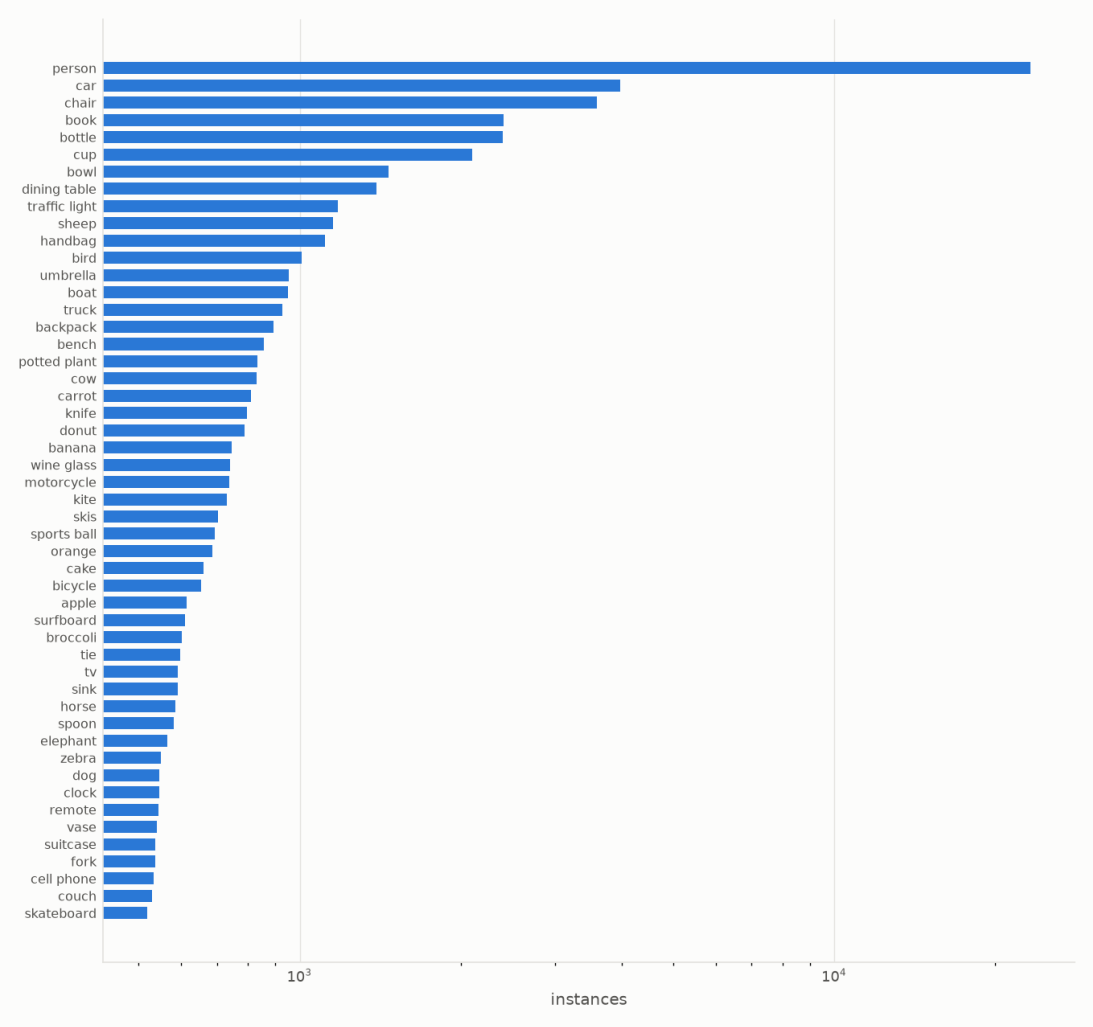


Figure 1: The 50 most frequent classes in the KB. Values are stored in `stats.json`, with person having 23,286 instances and appearing in 5,599 images. Other prominent classes include car, chair, book, bottle, cup, bowl, dining table, traffic light, and sheep.

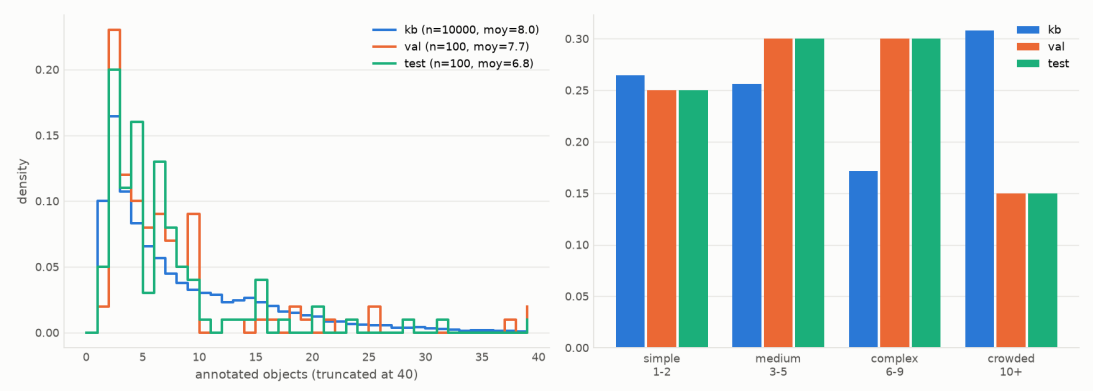


Figure 2: Scene-complexity distributions. The KB follows a natural distribution (26.47% simple, 25.61% medium, 17.15% complex, and 30.77% crowded), whereas validation and test follow the implemented 25/30/30/15% quotas.

at equal total frequencies, as do above/below and inside/contains. In the KB, the proportions are 28.668% for left/right, 10.412% for above/below, 10.351% for overlapping, 4.758% for inside/contains, and 1.973% for on-top-of; note that each inverse relation pair represents one stored relation in each directional interpretation. The same pattern holds across evaluation splits.

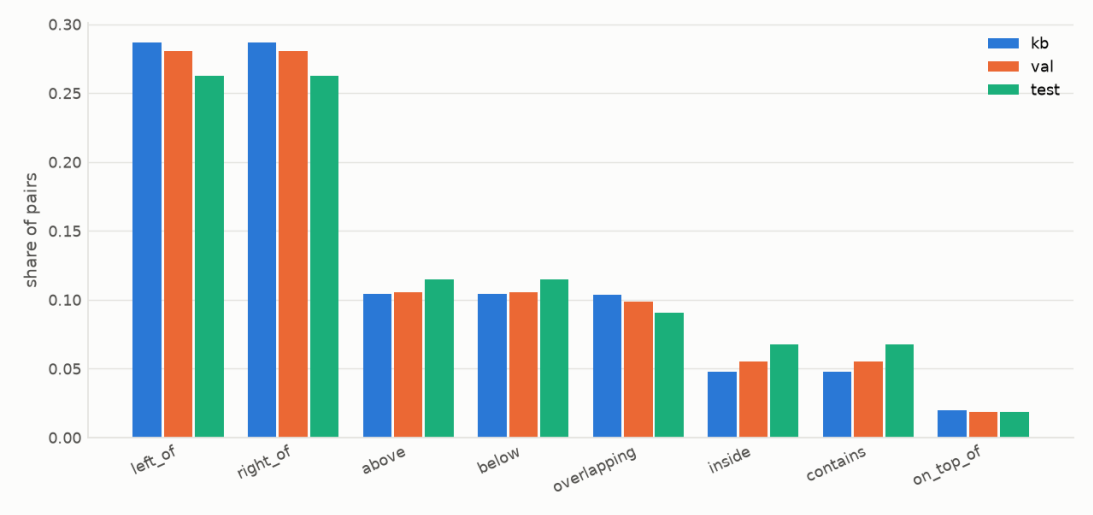


Figure 3: Spatial relations inferred from object bounding boxes. The symmetry between directional relation pairs is geometric by construction.

4.3 Captions and lexical statistics

Mean caption length is stable across splits: 10.483 KB tokens, 10.488 validation tokens, and 10.472 test tokens, with median 10 in every split. The full KB vocabulary contains 9,697 unique tokens, 3,936 hapax legomena, and 524,149 tokens. Token-level type-token ratios should not be compared directly across split sizes, because the value for KB is 0.0185, versus 0.1844 and 0.1765 for validation and test. Validation has 967 vocabulary items and the test has 924. Out-of-vocabulary rates relative to the KB are 1.09% for validation and 0.74% for test.

The repository compares KB and test caption lengths using a two-sample Kolmogorov-Smirnov statistic of $D = 0.02484$, $p = 0.9125$. This result supports the descriptive observation that caption lengths are similar despite the intentionally different stratification. Figure 4 combines caption-length and Zipf-style evidence.

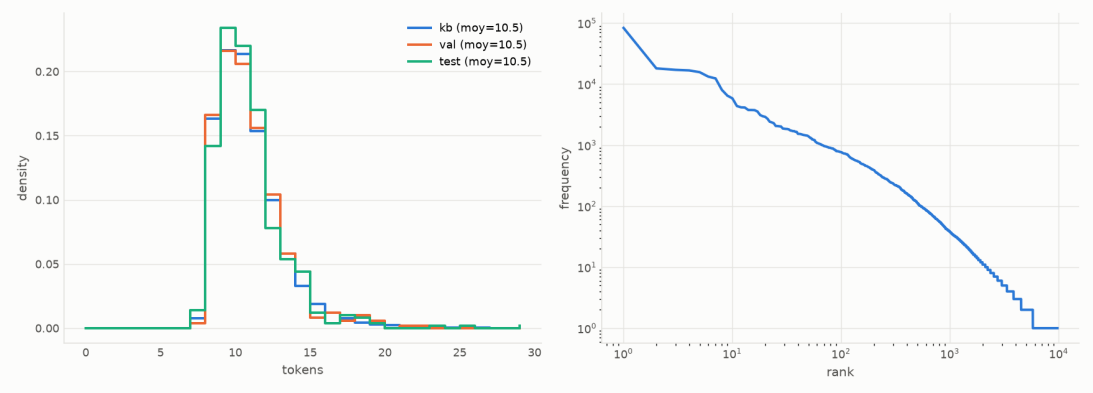


Figure 4: Caption-length distributions and associated corpus-frequency visualization. The short caption bodies are concentrated around the same median across all splits.

The NLTK part-of-speech distribution recorded for the KB has 74,604 nouns, 44,638 determiners, 33,596 prepositions, and 24,269 verbs, with lower counts for adjectives, pronouns, conjunctions, numerals, adverbs, particles, and unknown tags. Figure 5 displays the distribution.

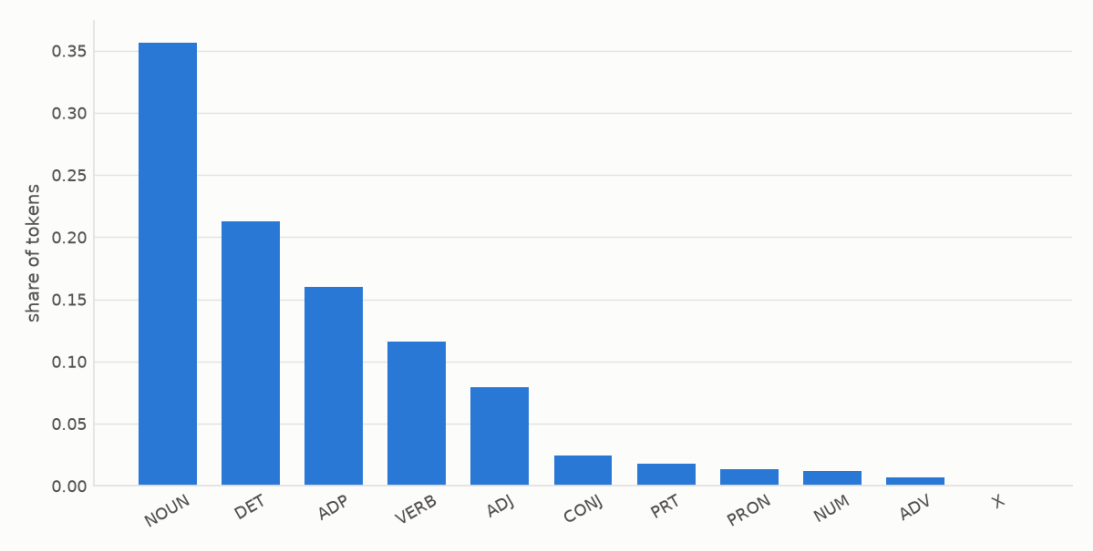


Figure 5: Part-of-speech distribution derived from the KB captions.

4.4 Co-occurrence and image-text association

The repository analyses frequent class co-occurrence and the association between object count and caption length. The recorded Spearman correlation in the KB is $\rho = 0.13858$, with $p = 4.5492 \times 10^{-44}$. Because the sample contains 50,000 captions, the correlation is statistically distinguishable from zero despite its small descriptive magnitude. It should not be interpreted as a direct causal relationship between scene object count and linguistic richness.

Figure 6 visualizes the leading class pairs. Figure 7 displays the relationship between annotation count and caption length.

4.5 Split representativeness

The Jensen-Shannon divergence of class distributions is 0.0311 for KB versus validation and 0.0313 for KB versus test, in a convention where zero denotes identical distributions. These small divergences indicate broadly similar class populations. Nevertheless, they do not demonstrate visual equivalence: validation and test are intentionally constrained samples, and their allocation differs from the KB's natural distribution.

All 80 classes occur in every split and the test has no missing class. Thus class coverage is complete at the category level, although the number of images representing each class in an evaluation split may be modest. Figure 8 summarizes context, illumination, object size, and related diversity measures.

In the KB, 92.16% of images are day/bright; in validation and test the corresponding shares are 97% and 91%. Gray is the leading dominant image colour in all splits. KB object-size shares are approximately 60.95% small, 26.42% medium, and 12.64% large. Figure 9 presents the colour distribution and Figure 10 provides the qualitative image sample grid.

Finally, the fixed caption attribute lexicon identifies colour, size, material, state, and shape expressions in KB captions. Its largest recorded category counts are 10,586 colour, 5,604 size, 2,532 state, 1,766 material, and 239 shape attributes. White is the most frequent colour attribute with 2,943 occurrences, followed by large and small. Shape counts include flat 73, round 57, square 53, circular 14, rectangular 12, and oval 9 occurrences in the figure's source data. Figure 11 illustrates these distributions.

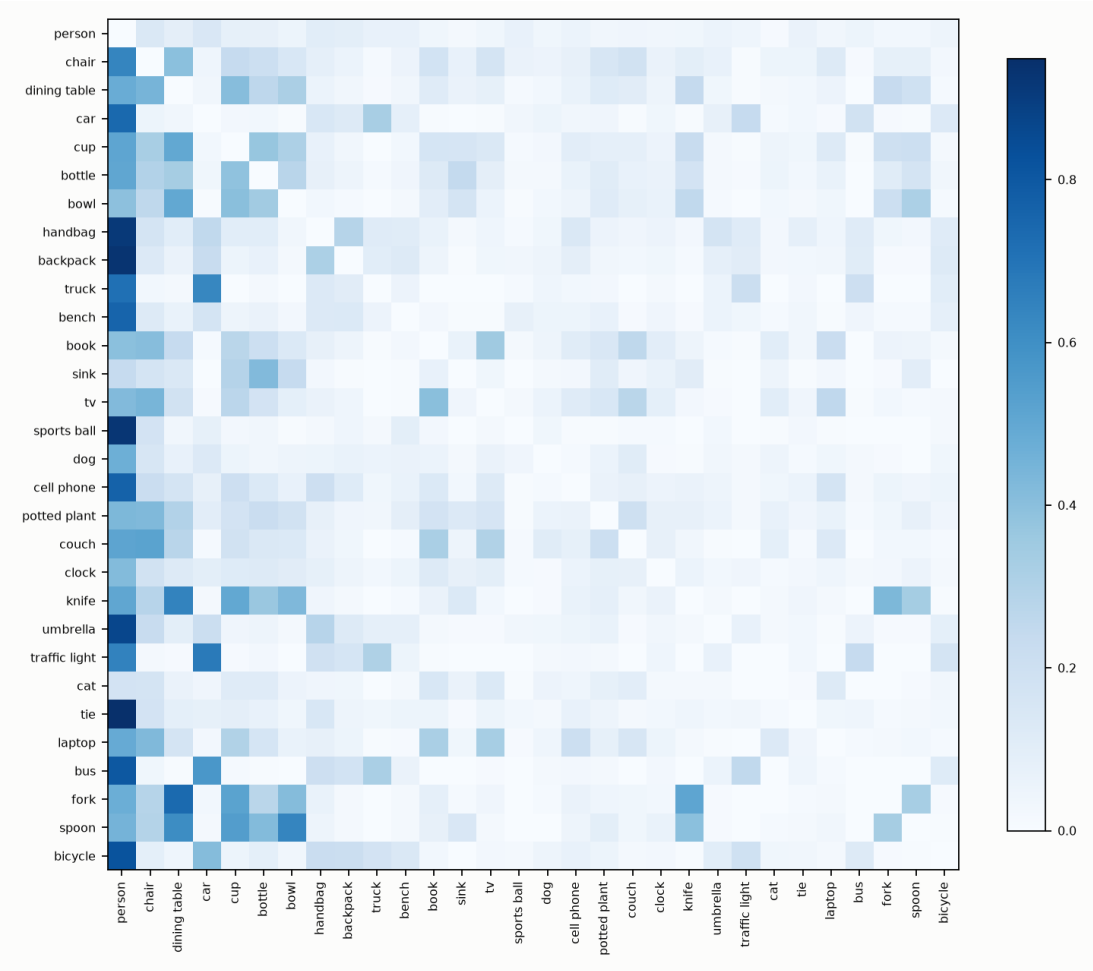


Figure 6: Top 30 object-class co-occurrences in the KB.

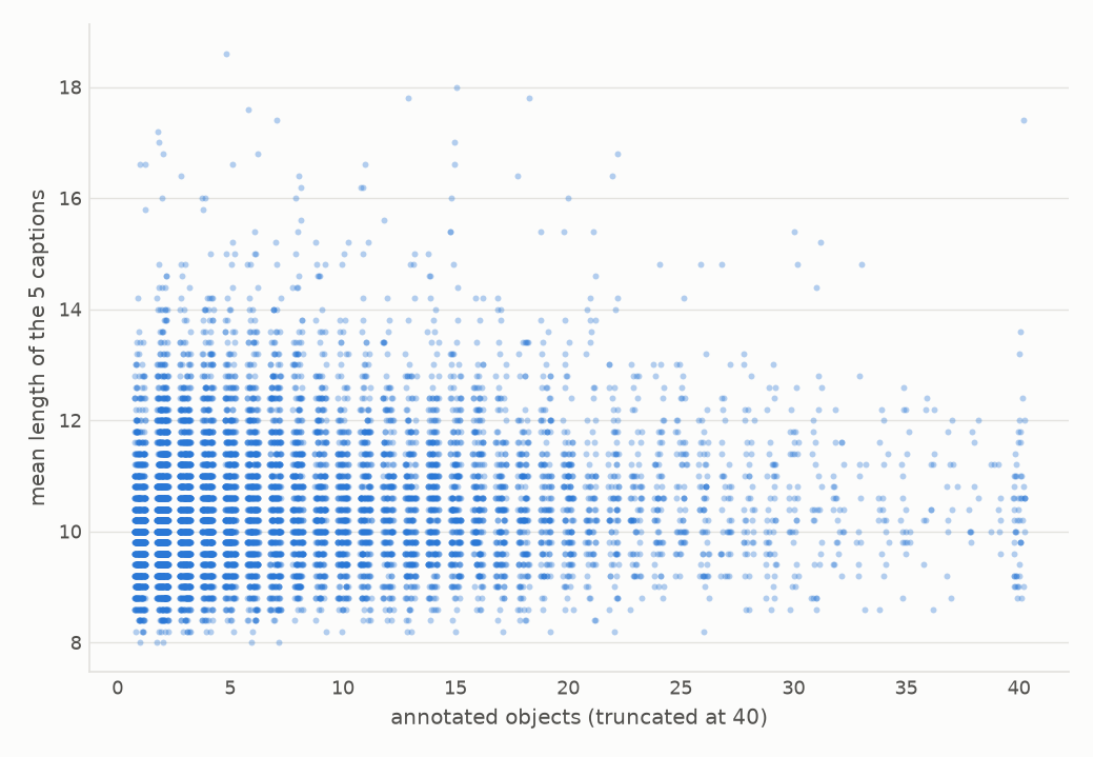


Figure 7: Relationship between object count and caption length.

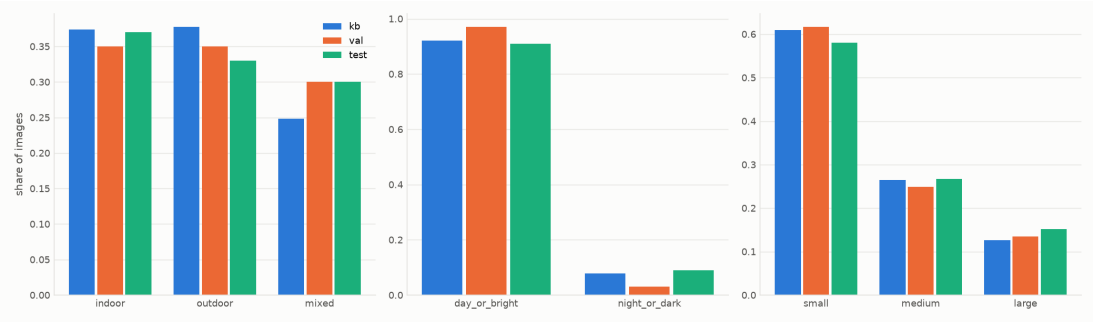


Figure 8: Context and visual-diversity distributions across splits.

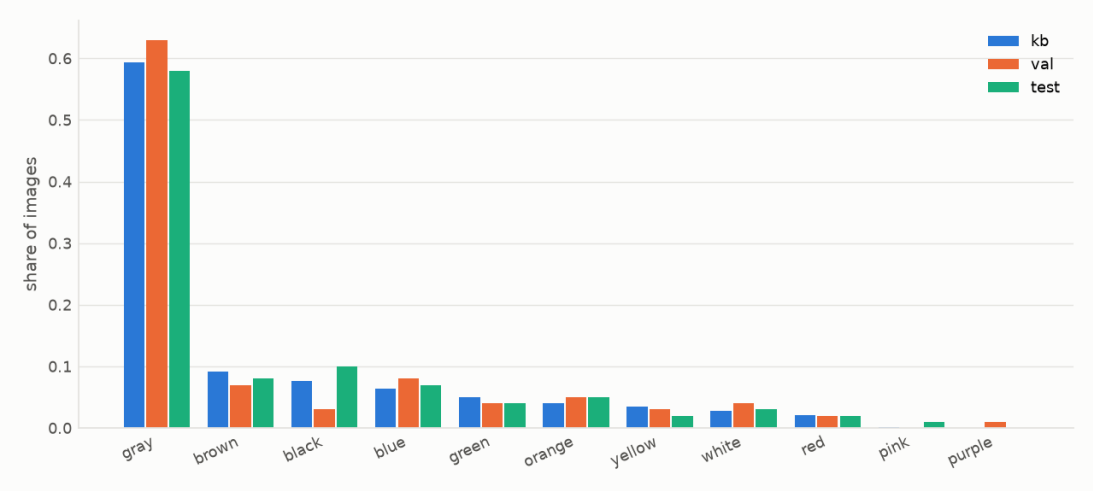


Figure 9: Dominant visual colours by split.

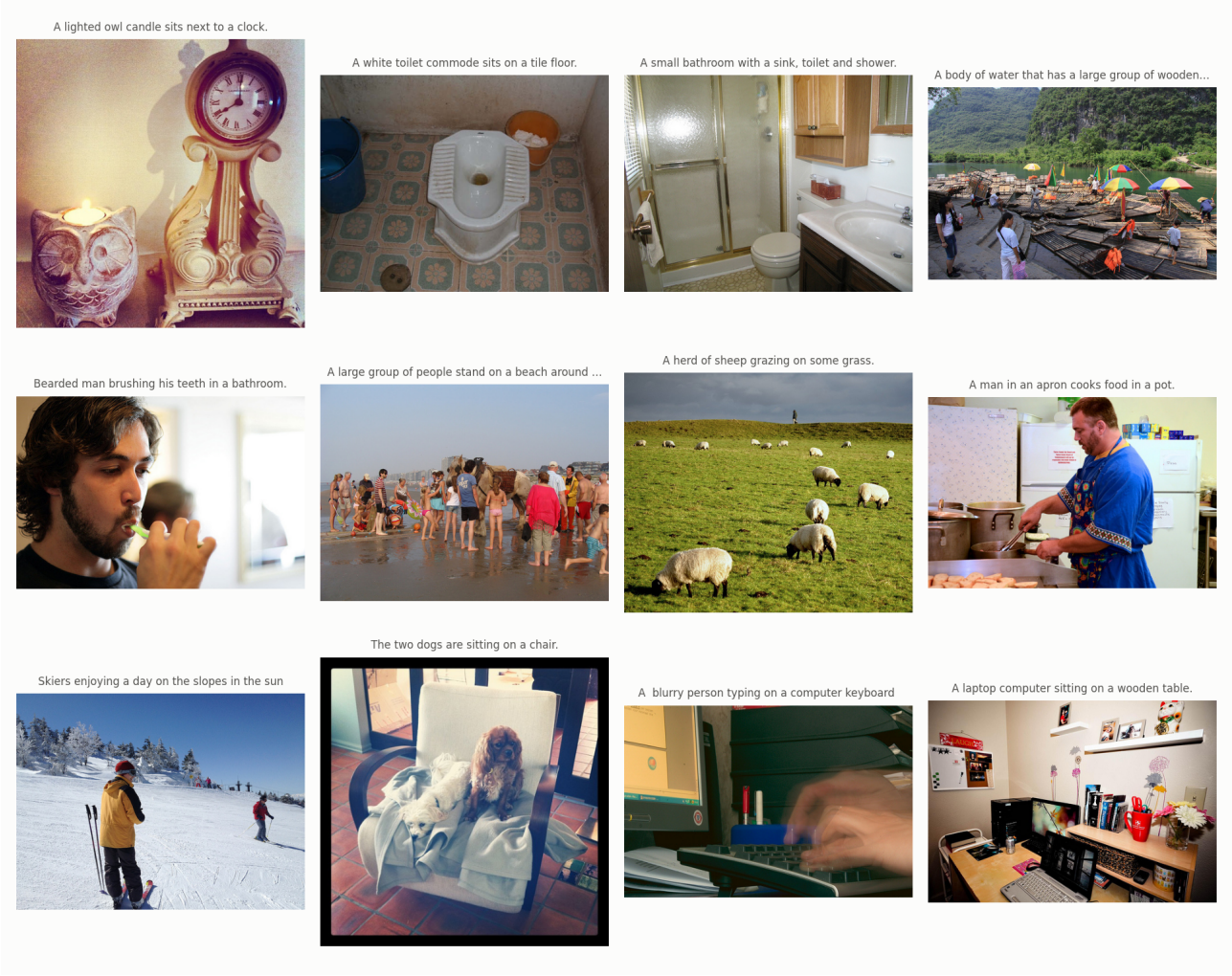


Figure 10: Qualitative sample from the frozen release.

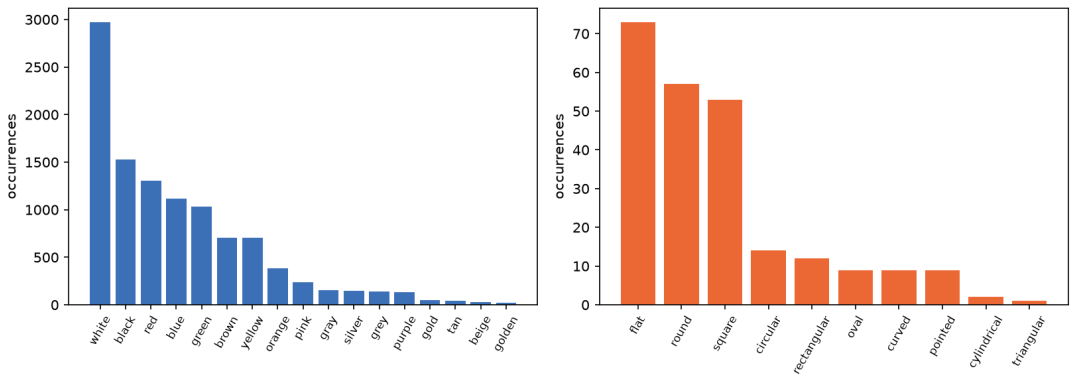


Figure 11: Distribution of lexical colour and shape attributes in KB captions.

5 Image and Caption Preprocessing

5.1 Image normalization

Each image is checked for validity and converted to RGB format. Images larger than 512 pixels on their longest side are resized while preserving their aspect ratio. Smaller images are not enlarged.

Images are saved as JPEG files with quality 95. The pipeline also records image dimensions, colour information, brightness, sharpness, and file hashes for quality control.

5.2 Model-specific preprocessing

Each model uses its own preprocessing tools. CLIP and SigLIP use their Hugging Face processors to prepare images and text, while DINOv2 uses its image processor and CLS token to extract image features.

CLIP uses a maximum text length of 77 tokens, while SigLIP uses 64 tokens.

5.3 Caption cleaning

For each KB image, the first five captions are selected and stored. The raw captions are preserved, while cleaned versions are converted to lowercase, stripped of unnecessary characters, and normalized for whitespace.

Evaluation captions remain separate from the KB to prevent evaluation references from being included in retrieval data

6 Metadata Extraction

Each split is joined to COCO instance annotations and caption records. Crowd annotations are ignored. For every non-crowd object, the pipeline records COCO category identity and supercategory, normalized bounding-box coordinates, image-area fraction, and size:

$$\text{small} : a < 0.02, \quad \text{medium} : 0.02 \leq a < 0.15, \quad \text{large} : a \geq 0.15,$$

where a is the object bounding-box area fraction after clipping to image bounds. For the largest objects, the pipeline estimates a named pixel colour for objects with area fraction at least 0.01, processing up to eight objects.

Top-level metadata also includes category counts, object counts, supercategories, complexity, context, day/night status, brightness, colorfulness, dominant colours, relations, and KB caption attributes. Complexity follows the thresholds in Section 3. Context is assigned through indoor/outdoor COCO supercategories. Day/night is based on the configured brightness threshold of 0.28.

Eight geometric relations—‘inside‘, ‘contains‘, ‘on_top_of‘, ‘overlapping‘, ‘left_of‘, ‘right_of‘, ‘above‘, and ‘below‘—are calculated from bounding-box geometry among the largest objects. They are not inferred by a semantic scene graph. Caption attributes come from fixed lexicons of 17 colours, 8 materials, 10 size words, 13 states, and shape expressions.

For implementation realism, the repository’s optional DETR path can supply predicted object detections. However, frozen validation/test objects in ‘metadata.jsonl‘ are COCO ground truth and therefore constitute an oracle for evaluating query metadata. This limitation is recorded explicitly and must be considered when interpreting downstream metadata-based results.

7 Data Quality Assessment and Anti-Leakage Measures

7.1 Image audit and repair

The image audit confirms that all 10,200 images are readable and valid. It also checks image formats, exact duplicates, and near-duplicates.

Two duplicate images were identified in the KB: one exact duplicate and one near-duplicate. Both were removed and replaced with other images. The final audit found no remaining duplicates or qualifying near-duplicates under the configured thresholds.

7.2 Cross-split identity controls

The three splits contain separate image IDs, and their integrity is verified using SHA-256 hashes. The retrieval API also prevents a query image from being returned as its own result.

Additional SigLIP similarity and difference-hash checks found no qualifying near-duplicates between the evaluation sets and the KB.

Table 7.2 summarizes the QC programme rather than reproducing all 40 individual rows.

Table 4. Consolidated quality-control outcome.

QC area	Result
Split identity and counts	40 checks executed; 40 passed
Invalid images	0
Exact duplicates	0
Validation/test near-duplicates	0/100 in each split using SigLIP cosine and dHash
KB caption coverage	Exactly 5 caption rows/image
Evaluation reference coverage	Exactly 5 rows/image
Embeddings and indexes	Sizes aligned to split and embedding dimensions
Retrieval smoke tests	Passed for validation and test

Table 4 should be read together with the detailed repository audit. The frozen report contains 40 checks, no failures, and no warnings.

7.3 Caption-string overlap

The QC normalized evaluation captions and KB captions with the same cleaning function. It found normalized string matches on 5 of 500 validation reference rows (1.0%) and 7 of 500 test reference rows (1.4%). Those matches involved 9 and 37 evaluation-image-to-KB-image mappings, respectively, because generic descriptions occur on multiple KB images.

This result is not interpreted as image leakage. Every listed pair has different evaluation and KB image IDs, and clean-string equality alone does not establish that either the underlying image or a source caption record leaked across splits. The audit explicitly reports these occurrences so that repetitive generic captions do not conceal their presence.

8 Image and Text Embeddings

The pipeline generates image and text embeddings using four models: CLIP [5], SigLIP [8], DINOv2 [4], and Sentence-BERT [6]. These embeddings represent images and captions as numerical vectors that can be compared using similarity measures.

All embeddings are stored as normalized float32 vectors. Table 5 summarizes the dimensions and number of vectors generated by each model.

Table 5. Embedding products and frozen shapes.

Product	Image KB	Val/test images	Text KB
CLIP ViT-B/32	$10{,}000 \times 512$	100×512	$50{,}000 \times 512$
SigLIP base patch16/224	$10{,}000 \times 768$	100×768	$50{,}000 \times 768$
DINOv2-small	$10{,}000 \times 384$	100×384	—
SBERT all-MiniLM-L6-v2	—	—	$50{,}000 \times 384$

Table 5 demonstrates aligned vector dimensions within each image/text space and fixed row alignment with image and caption IDs. CLIP and SigLIP support both image-to-image and image-to-text retrieval. DINOv2 supports only image vectors in the frozen release, while SBERT supplies a separate caption embedding not linked to a FAISS index in this release.

For vectors $\mathbf{x}, \mathbf{y}$, normalization permits cosine similarity to be computed as

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} \approx \mathbf{x}^\top \mathbf{y}, \qquad \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1.$$

The consequence is that a Euclidean inner-product index on normalized vectors implements cosine retrieval.

9 Knowledge Base and FAISS Index Construction

9.1 Exact image and text indexes

FAISS [2] is used to build similarity-search indexes for the KB. The image indexes contain 10,000 vectors each, while the CLIP and SigLIP text indexes contain 50,000 caption vectors each.

The indexes use IndexFlatIP, which performs exact inner-product search. Since the vectors are normalized, this is equivalent to cosine similarity.

Table 9.1 summarizes the index configurations and their build times.

Table 6. Image FAISS index configurations.

Encoder	Dimension	Flat size	Flat build	HNSW size	HNSW build	Parameters
CLIP	512	20.480 MB	0.0081 s	23.198 MB	4.0060 s	$M = 32$, efConstruction 200, efSearch 64
DINOv2	384	15.360 MB	0.0497 s	18.078 MB	6.8016 s	$M = 32$, efConstruction 200, efSearch 64
SigLIP	768	30.720 MB	0.0225 s	33.438 MB	1.4390 s	$M = 32$, efConstruction 200, efSearch 64

Table 6 makes clear that storage grows primarily with vector dimension rather than the same number of vectors multiplied by byte width. The flat image-index build measurements are reported by ‘index_report.json‘; however, they are not comparable to the separate NumPy latency benchmark used in encoder comparison scripts.

The CLIP and SigLIP caption indexes are 512- and 768-dimensional ‘IndexFlatIP‘ structures containing 50,000 vectors. Filesystem sizes are 102,400,045 bytes and 153,600,045 bytes, respectively. The index report does not record separate text-index build times or MB values, so no value is inferred for those fields.

9.2 Approximate HNSW search

HNSW indexes provide faster approximate similarity search. They use M=32, efConstruction=200, and efSearch=64.

Recall@10 is 1.000 for CLIP, 0.997 for DINOv2, and 0.999 for SigLIP, showing that approximate search closely matches exact search. For SigLIP, median search latency decreases from 0.737 ms with flat search to 0.082 ms with HNSW.

9.3 Retrieval API

The public ‘Release‘ class loads the configured encoder, performs image and caption retrieval, and maps returned KB IDs to raw captions. Flat and HNSW image indexes are separately addressable. A query with supplied ‘query_ids‘ is checked against every returned ID and raises an error if its own image is returned. For the image-to-text mode, query vectors are searched against the 50,000 caption vectors and aligned parent-image IDs are returned.

10 Encoder Comparison and Validation-Based Selection

10.1 Retrieval configurations

Five retrieval configurations are compared using CLIP, DINOv2, and SigLIP.

In image-to-image retrieval, the query image is compared with KB images, and the caption of the most similar image is returned. In image-to-text retrieval, the query image is compared directly with KB caption embeddings to retrieve the most similar caption.

The configurations are evaluated using retrieval metrics and CIDEr [7]. A random-caption baseline is also included for comparison.

10.2 Validation selection metric

CIDEr [7] is used as the main metric for selecting the best configuration. Each configuration is evaluated on the 100 validation images, and the one with the highest mean CIDEr is selected. Jaccard class@1 is used as a tie-breaker.

SigLIP image-to-text achieves the highest validation CIDEr of 0.7223 and is therefore selected. The test set is kept separate and is used only for final reporting.

$$\text{selected} = \arg\max_{c} \frac{1}{100} \sum_{i=1}^{100} \text{CIDEr}(c, i).$$

The selection was made before looking at the final test results. Test data were retained in Table 8 and were not included in configuration selection.

11 Experimental Results and Statistical Analysis

11.1 Validation results

Table 7 gives the validation metrics recorded in the comparison artifact and repository CSV.

Table 7. Validation configuration comparison.

Configuration	Recall@1	Recall@5/@10	Precision@1/@5/@10	MRR	Jaccard@1	CIDEr
SigLIP image-to-text	0.96	1.00 / 1.00	0.96 / 0.972 / 0.962	0.9775	0.4994	**0.7223**
SigLIP image-to-image	1.00	1.00 / 1.00	1.00 / 0.976 / 0.969	1.0000	0.5455	0.6735
CLIP image-to-text	0.97	1.00 / 1.00	0.97 / 0.972 / 0.970	0.9850	0.5349	0.5584
DINOv2 image-to-image	0.98	1.00 / 1.00	0.98 / 0.960 / 0.961	0.9900	0.5209	0.5541
CLIP image-to-image	0.98	1.00 / 1.00	0.98 / 0.972 / 0.951	0.9883	0.5382	0.5286

Table 7 shows that SigLIP image-to-text achieves the highest validation CIDEr score of 0.7223. However, SigLIP image-to-image achieves better recall@1 and MRR. This shows that retrieval accuracy and caption quality do not always produce the same ranking.

The reported confidence intervals, mean differences, and Wilcoxon comparisons against SigLIP image-to-text are reproduced in Table 8.

Table 8. Validation statistical comparison with SigLIP image-to-text.

Configuration	Mean CIDEr	95% CI	Difference vs. best	Wilcoxon p
SigLIP image-to-text	0.7223	[0.6168, 0.8387]	—	—
SigLIP image-to-image	0.6735	[0.5753, 0.7825]	0.0487	0.3023
CLIP image-to-text	0.5584	[0.4626, 0.6585]	0.1638	0.00405
DINOv2 image-to-image	0.5541	[0.4695, 0.6516]	0.1682	0.01117
CLIP image-to-image	0.5286	[0.4384, 0.6233]	0.1937	0.00915

Table 8 presents the statistical comparison between the selected configuration and the alternatives. SigLIP image-to-image is not significantly different from SigLIP image-to-text ($p = 0.3023$). The other three configurations are significantly lower under the Bonferroni threshold of 0.0125 [1].

Therefore, SigLIP image-to-text is selected based on its highest validation CIDEr, although its advantage over SigLIP image-to-image is not statistically significant.

11.2 Final test reporting

Table 11.2 presents report-only test values. These results are not used to select a configuration.

Table 9. Report-only test results.

Configuration	Recall@1	Recall@5/@10	Precision@1/@5/@10	MRR	CIDEr	Random CIDEr
CLIP image-to-image	0.93	1.00 / 1.00	0.93 / 0.926 / 0.905	0.9567	0.6118	0.0420
CLIP image-to-text	0.96	1.00 / 1.00	0.96 / 0.960 / 0.957	0.9770	0.6501	0.0420
DINOv2 image-to-image	0.98	1.00 / 1.00	0.98 / 0.948 / 0.951	0.9870	0.5503	0.0420
SigLIP image-to-image	0.95	0.99 / 1.00	0.95 / 0.956 / 0.953	0.9717	0.6216	0.0420
SigLIP image-to-text	0.98	1.00 / 1.00	0.98 / 0.964 / 0.972	0.9883	**0.7064**	0.0420

Table 9 presents the final test results. SigLIP image-to-text achieves the highest CIDEr score of 0.7064, followed by CLIP image-to-text at 0.6501. The random-caption baseline scores 0.0420.

These results are used only for final reporting and do not influence the configuration selected using validation data.

Figure 12 visualizes validation CIDEr by configuration. The repository has one encoder-comparison chart, named ‘encoder_comparison_val.png‘.

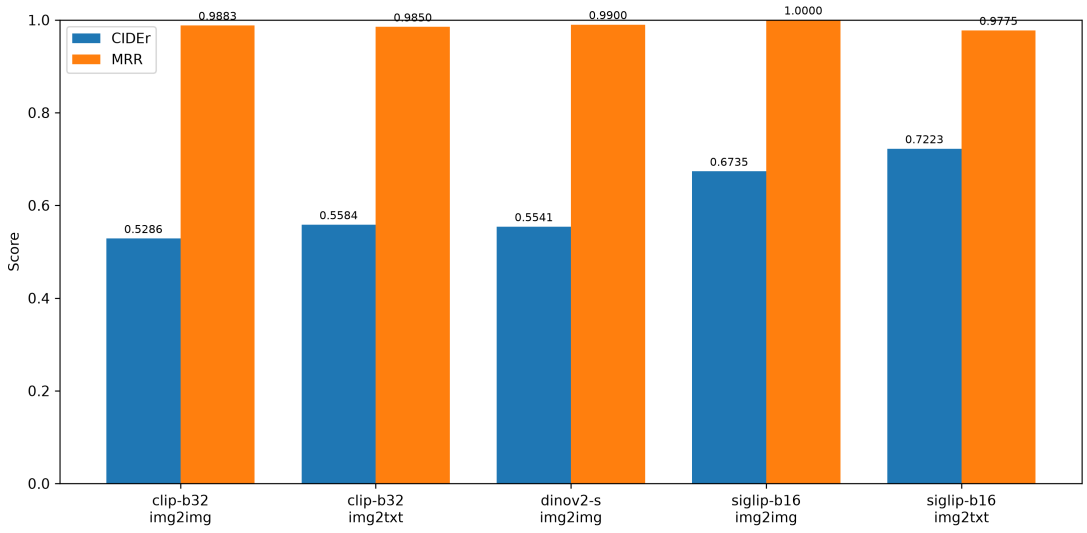


Figure 12: Validation CIDEr comparison for the five implemented retrieval configurations.

11.3 Interpretation

Several conclusions are statistically and descriptively supported, with qualifications. SigLIP image-to-text is the validation winner and final test leader. SigLIP image-to-image is not statistically distinguishable from the winner on validation data, so the empirical evidence does not justify treating the 0.0487 validation CIDEr difference as established. CLIP image-to-text, DINOv2 image-to-image, and CLIP image-to-image are each significantly below SigLIP image-to-text under the repository’s Bonferroni threshold. A random caption baseline remains substantially lower.

No claim is made that the comparison establishes general superiority of SigLIP across all retrieval settings. The sample has only 100 validation images, 100 final test images, one run per configuration in the frozen artifacts,

and a nearest-neighbour caption evaluation protocol. Positive test consistency is supportive but not sufficient for broad generalization.

12 Reproducibility and Dataset Freezing

12.1 Configuration and execution

The master configuration records seed 42, release version g1-v1.1, split sizes, caption cardinality, sampling thresholds, image normalization parameters, encoder identifiers, and FAISS settings. Split seeds are 43, 44, and 45 for test, validation, and KB respectively within the pipeline's fixed base seed. The recommended full notebook order begins with annotations and ends with the retrieval demonstration; a dedicated runner exists for frozen SigLIP extraction, QC, freeze verification, synthetic mini-COCO testing, and cached encoder checks.

The frozen runtime recorded by 'artifacts/manifest.json' is Python 3.13.7 on Windows 11. Package versions include NumPy 2.5.4, pandas 3.1.0, SciPy 1.18.1, Pillow 13.0.0, Matplotlib 3.11.2, FAISS 1.15.1, PyTorch 2.15.0 CPU, Transformers 5.22.1, and Sentence-Transformers 6.2.0. 'requirements.txt' also pins scikit-learn 1.9.1, COCO evaluation tools, sentencepiece, protobuf, imagehash, Jupyter packages, and supporting dependencies.

12.2 Manifest and read-only release

The manifest identifies release g1-v1.1, seed 42, primary encoder SigLIP, and 10,270 hashed files. Frozen artifacts are subsequently set read-only. 'run_freeze_only.py –verify' checks the manifest SHA-256 values and is documented to print "release identique à la référence" for an intact release.

This manifest guarantees file identity for users receiving the same release. It does not guarantee bitwise identical embeddings when they are regenerated on a different device, especially where hardware, kernels, or dependency behavior changes floating-point results by approximately 10^{-6}. It also does not claim that run time remains identical.

The repository preserves historical provenance: manifest v1.0 records CLIP and DINOv2 runtime under Python 3.11.9, PyTorch 2.12.0 CPU, and Transformers 4.57.6. One documentation note for SigLIP says Python 3.13.3, while 'manifest.json' records 3.13.7; the report therefore treats the manifest value as the frozen runtime and records the discrepancy as a known documentation limitation.

12.3 Dataset freezing

The freeze separates raw input integrity from generated artifact integrity. Split files include SHA-256 hashes of ID lists, while the manifest hashes file content. Verification is performed through content hashes rather than only file names or counts. Any changes to images, embeddings, indexes, splits, or QC artifacts require rebuilding a new release rather than modifying the frozen directory.

13 Limitations and Conclusion

13.1 Limitations

Phase 1 does not include a complete caption-generation model. It evaluates retrieval-based caption candidates instead.

The validation and test sets contain only 100 images each, which limits the reliability and generalizability of the results.

Validation and test metadata use COCO ground-truth annotations rather than predicted object detections. The geometric relationships and caption attributes are also based on predefined rules and cannot fully represent semantic relationships.

Finally, embedding reproducibility may vary across hardware and software environments. The frozen release guarantees file integrity, but not identical execution times or bitwise-identical regenerated embeddings.

13.2 Conclusion

Phase 1 successfully establishes a shared dataset and retrieval infrastructure for future image-captioning research.

The release contains 10,200 images, 51,000 captions, extracted metadata, image and text embeddings, FAISS indexes, and quality-control checks. It also includes five retrieval configurations evaluated using validation and test data.

SigLIP image-to-text achieves the highest validation CIDEr of 0.7223 and the highest report-only test CIDEr of 0.7064. It is therefore selected as the best-performing configuration in this phase, although its validation advantage over SigLIP image-to-image is not statistically significant.

The resulting dataset and retrieval baseline provide the foundation for Phase 2, which will focus on implementing the final caption-generation system.

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